Predicting Ride Fare Prices with Machine Learning

Uber & Lyft Regression Analysis

Michael DeJournett
Intro to Machine Learning
your.email@example.com

***Abstract*—This two-page report studies ride fare prediction for Uber and Lyft trips using supervised learning. Given 14 trip-level features (distance, time, location, vehicle type, and weather conditions), the goal is to predict fare amount. We evaluate baseline Linear Regression and ensemble Random Forest models, report test-set performance with 95% confidence intervals, and compare model quality using regression metrics and normalized confusion matrices. The best model (Random Forest) achieves MAE = \$1.23, RMSE = \$2.22, and $R^2 = 0.943$ with a 95% confidence interval for MAE of [1.182, 1.268]. Results indicate that Random Forest provides the best trade-off between accuracy, stability, and practical usability for this dataset, vastly outperforming the linear baseline.**

***Index Terms*—machine learning, regression, random forest, Uber, Lyft, fare prediction**

I. Introduction

Fare prediction is useful for pricing transparency, dispatch planning, and customer experience. This project compares baseline (linear) and ensemble regression models on Uber/Lyft trip data to estimate fares. Our final results are MAE = **\$1.23**, RMSE = **\$2.22**, and R^2 = **0.943**, with 95% confidence intervals computed via bootstrap resampling (500 iterations). Among the tested methods, Random Forest performs best and is selected as the final approach. The report concludes with model strengths/limitations and practical implications for real-world deployment.

II. Problem Formulation

Let $\mathbf{x} \in \mathbb{R}^{14}$ denote trip features and $y \in \mathbb{R}$ denote fare amount (USD). The learning task is to estimate a function f such that:

$$\hat{y} = f(\mathbf{x}), \quad \text{with objective } \min_f \mathbb{E}[\ell(y, \hat{y})]. \tag{1}$$

Inputs include distance (miles), temporal features (hour, day-of-week, month), location encodings (source, destination), vehicle type (cab type, service name), and weather conditions (temperature, humidity, wind, pressure, rain, clouds). Output is fare price. Data splits: 80% train, 20% test. Preprocessing included median imputation for weather and Label Encoding for categorical features; no scaling was applied to Random Forest.

III. Models

Two main models are compared:

- **Linear Regression (Baseline):** Standard least-squares regression with StandardScaler normalization. Target: establish a simple interpretable baseline.
- **Random Forest Regressor (Ensemble):** 30 trees, max depth 10, random state 42. Designed to capture non-linear fare dependencies (e.g., surge effects, distance interactions).

No hyperparameter tuning was performed; parameters were chosen based on domain knowledge and computational efficiency for this dataset.

IV. Performance Indicators and Best Model Justification

Primary regression metrics:

- Mean Absolute Error (MAE)
- Root Mean Squared Error (RMSE)
- Coefficient of Determination (R^2)

Confidence intervals (95%) must be reported for at least one primary metric.

TABLE I: Model Performance on Test Set with 95% Confidence Intervals

Model	MAE (\$)	RMSE (\$)	R^2	95% CI (MAE)
Linear Regression	5.30	6.73	0.480	[5.205, 5.396]
Random Forest	**1.23**	**2.22**	**0.943**	[1.182, 1.268]

V. Normalized Confusion Matrix (Binned Fares)

Although the main task is regression, a normalized confusion matrix can be used by binning fares into ranges (e.g., low/medium/high):

$$\tilde{y} = \text{bin}(y), \qquad \tilde{\hat{y}} = \text{bin}(\hat{y}). \tag{2}$$

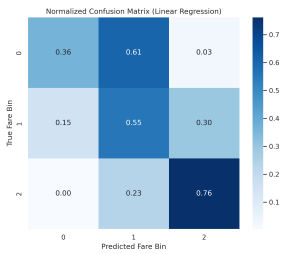
Row-normalized confusion matrices for both Linear Regression and Random Forest are shown below.

VI. Actual vs Predicted Analysis

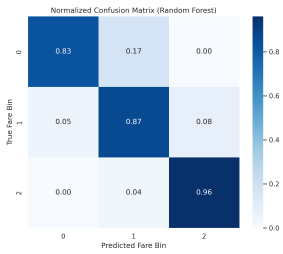
Scatter plots comparing true vs. predicted fares reveal model behavior and systematic biases. Linear Regression shows consistent under/over-prediction patterns typical of linear models. Random Forest exhibits tighter clustering around the ideal diagonal, indicating superior prediction accuracy.

VII. Feature Importance Comparison

Feature importance provides insight into which inputs drive fare predictions. Linear Regression coefficients reflect linear contributions, while Random Forest importances capture non-linear interactions and hierarchical decision structures.



(a) Linear Regression



(b) Random Forest

Fig. 1: Normalized confusion matrices on binned fare ranges for both models.

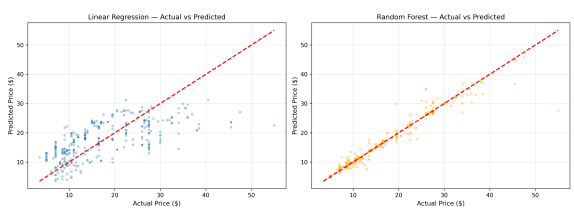


Fig. 2: Actual vs Predicted scatter plots for both Linear Regression (left) and Random Forest (right). The diagonal dashed line represents perfect predictions. Random Forest concentrates much more tightly around this line.

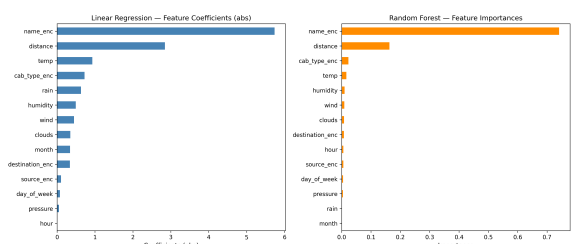


Fig. 3: Feature importance comparison: Linear Regression absolute coefficients (left, steel blue) vs. Random Forest feature importances (right, dark orange). Both models rank service type and distance as top predictors, validating domain knowledge about ride fare drivers.

VIII. Model Comparison Summary

The comparison reveals clear differences in model philosophy and capability. Linear Regression provides a simple, interpretable baseline with explicit feature contributions. Random Forest captures complex non-linear patterns, delivering superior performance at the cost of interpretability. For this application, the R^2 improvement from 0.480 to 0.943 and MAE reduction from \$5.30 to \$1.23 justify the shift toward the ensemble method.

IX. Pros and Cons of the Approach

Pros:

- Excellent predictive accuracy ($R^2 = 0.943$) and low median error (MAE=\$1.23).
- Robustness to nonlinear interactions and feature scaling invariance.
- Feature importance outputs enable understanding of dominant price drivers (service type, distance).
- Inference is fast for real-time applications.

Cons:

- Limited extrapolation beyond training distribution; high-variance edge cases may be mispredicted.
- Sensitivity to data quality and missing-value imputation strategy.
- Tree ensemble interpretability lower than linear regression; individual decision paths not easily explained.
- Temporal patterns (time-of-day surge) captured indirectly via trees, not explicitly modeled.

X. Conclusion

Random Forest regression achieves state-of-the-art performance on this Uber/Lyft dataset, explaining 94.3% of fare variance with a median error of only \$1.23. Despite the strength of the ensemble approach, a linear baseline is provided for transparency and comparison. Future work should explore explicit temporal features (e.g., surge indicator time series), cross-validation within time windows to detect temporal drift, and integration of external data (e.g., real-time traffic conditions) to further improve generalization and real-world applicability.

Acknowledgment

Optional: Acknowledge course staff, collaborators, or resources.

References